



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : Bachelor Of Information Technology (Hons) In Computer System Security
Course Code & Name : IKB21403 - VULNERABILITY ANALYSIS
Course Lecturer(s) : Nor Adani Kamal Mohamad Nasir
Submission Title : Lab-C5 Enumeration
Deadline : Day 10 Month 05 Year 2026 Time 11.59pm
Penalties :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

- | | |
|-------------------------------------|--|
| ☒ | I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations. |
| ☒ | This submission is my own, unless indicated with proper referencing. |
| ☒ | This submission has not been previously submitted or published. |
| ☒ | This submission follows the requirements stated in the course. |

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
	1 HAZIQ DANIAL BIN NOR AZAN	52215225213

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)

Introduction

Enumeration is the process of gathering detailed information from a target system. In cybersecurity, enumeration helps identify open ports, operating systems, users, services, protocols, and possible vulnerabilities. The purpose of this lab is to perform different enumeration techniques against a vulnerable machine using Kali Linux tools.

The victim machine used in this lab was Metasploitable2 running on VirtualBox. Kali Linux was used as the attacker machine.

Lab Environment

Component	Description
Attacker Machine	Kali Linux
Victim Machine	Metasploitable2
Network Type	VirtualBox Host-Only Network
Victim IP Address	192.168.56.105

Challenge 2 — Fast Nmap Scan

Objective

To quickly identify common open ports and services running on the target machine.

Command Used

```
nmap -F 192.168.56.105
```

Explanation of Command

- nmap = Network scanning tool.
- -F = Fast scan mode that scans the top 100 common ports.
- 192.168.56.105 = Target victim machine IP address.

```
(kali@kali)-[~]
$ nmap -F 192.168.56.105
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 13:48 -0400
Nmap scan report for 192.168.56.105
Host is up (0.0017s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 08:00:27:C7:24:F7 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds

(kali@kali)-[~]
$
```

Findings

The scan identified several open ports including:

- FTP (21)
- SSH (22)
- Telnet (23)
- HTTP (80)
- SMB (445)

Analysis

The target machine exposes multiple network services. These services may be vulnerable to attacks if outdated or misconfigured.

Challenge 5 — TTL OS Fingerprinting

Objective

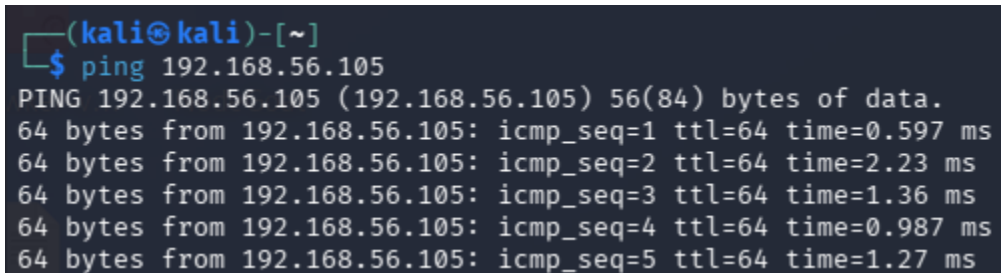
To identify the target operating system using TTL values.

Command Used

ping 192.168.56.105

Explanation of Command

- ping = Sends ICMP echo request packets.
- 192.168.56.105 = Target IP address.

A terminal window with a dark background. The prompt is (kali@kali)-[~]. The user has entered the command \$ ping 192.168.56.105. The output shows five successful ping responses, each with a TTL of 64.

```
(kali@kali)-[~]  
$ ping 192.168.56.105  
PING 192.168.56.105 (192.168.56.105) 56(84) bytes of data.  
64 bytes from 192.168.56.105: icmp_seq=1 ttl=64 time=0.597 ms  
64 bytes from 192.168.56.105: icmp_seq=2 ttl=64 time=2.23 ms  
64 bytes from 192.168.56.105: icmp_seq=3 ttl=64 time=1.36 ms  
64 bytes from 192.168.56.105: icmp_seq=4 ttl=64 time=0.987 ms  
64 bytes from 192.168.56.105: icmp_seq=5 ttl=64 time=1.27 ms
```

Findings

The ping response showed:

TTL=64

Analysis

TTL value 64 commonly indicates a Linux or Unix operating system. This suggests the target machine is running Linux.

Challenge 9 — FTP Banner Enumeration

Objective

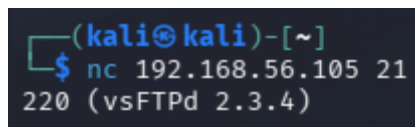
To identify the FTP service version running on the target machine.

Command Used

```
nc 192.168.56.105 21
```

Explanation of Command

- nc = Netcat tool used for network connections.
- 192.168.56.105 = Target IP address.
- 21 = FTP service port.



```
(kali㉿kali)-[~]  
$ nc 192.168.56.105 21  
220 (vsFTPd 2.3.4)
```

Findings

The FTP banner returned:

220 (vsFTPd 2.3.4)

Analysis

The target machine is running vsFTPd version 2.3.4. Older versions of FTP services may contain known vulnerabilities.

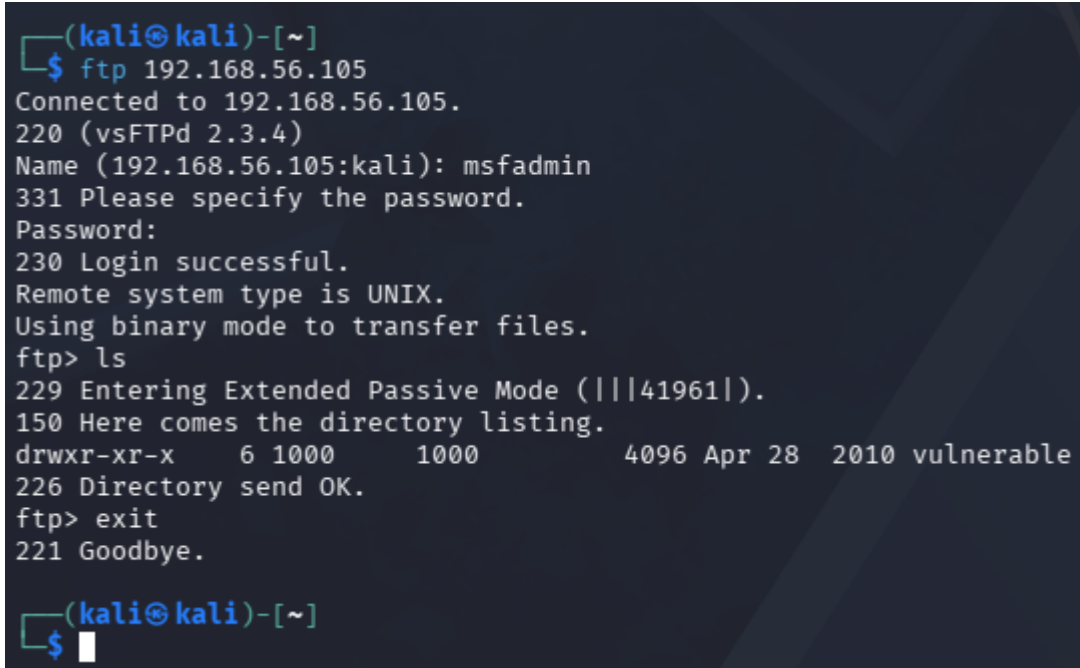
Challenge 10 — Anonymous FTP Login

Objective

To determine whether the FTP server allows login access and directory enumeration.

Command Used

ftp 192.168.56.105



```
(kali㉿kali)-[~]  
$ ftp 192.168.56.105  
Connected to 192.168.56.105.  
220 (vsFTPd 2.3.4)  
Name (192.168.56.105:kali): msfadmin  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> ls  
229 Entering Extended Passive Mode (|||41961|).  
150 Here comes the directory listing.  
drwxr-xr-x    6 1000    1000        4096 Apr 28  2010 vulnerable  
226 Directory send OK.  
ftp> exit  
221 Goodbye.  
  
(kali㉿kali)-[~]  
$
```

Findings

The FTP login was successful using credentials:

Username: msfadmin

Password: msfadmin

Directory listing showed:

vulnerable

Analysis

The FTP server allows authenticated access and exposes accessible directories. Attackers may use FTP enumeration to identify sensitive files or upload malicious content.

Challenge 11 — SMB NSE Enumeration

Objective

To enumerate SMB information using Nmap NSE scripts.

Commands Used

```
nmap --script smb-os-discovery -p445 192.168.56.105
```

```
nmap --script smb-enum-users -p445 192.168.56.105
```

Explanation of Commands

- --script smb-os-discovery = Detect SMB operating system information.
- --script smb-enum-users = Enumerate SMB users.
- -p445 = Scan SMB port.

```
(kali㉿kali)-[~]  
└─$ nmap --script smb-os-discovery -p445 192.168.56.105  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 13:53 -0400  
Nmap scan report for 192.168.56.105  
Host is up (0.00047s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:C7:24:F7 (Oracle VirtualBox virtual NIC)  
  
Host script results:  
| smb-os-discovery:  
|   OS: Unix (Samba 3.0.20-Debian)  
|   Computer name: metasploitable  
|   NetBIOS computer name:  
|   Domain name: localdomain  
|   FQDN: metasploitable.localdomain  
|_  System time: 2026-05-06T13:53:43-04:00  
  
Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

```

(kali@kali)-[~]
$ nmap --script smb-enum-users -p445 192.168.56.105
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 13:54 -0400
Nmap scan report for 192.168.56.105
Host is up (0.00077s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:C7:24:F7 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-enum-users:
|   METASPLOITABLE\backup (RID: 1068)
|     Full name: backup
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\bin (RID: 1004)
|     Full name: bin
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\bind (RID: 1210)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\daemon (RID: 1002)
|     Full name: daemon
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\dhcp (RID: 1202)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\distccd (RID: 1222)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\ftp (RID: 1214)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\games (RID: 1010)
|     Full name: games
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\gnats (RID: 1082)
|     Full name: Gnats Bug-Reporting System (admin)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\irc (RID: 1078)
|     Full name: ircd
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\klog (RID: 1206)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\libuuid (RID: 1200)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\list (RID: 1076)
|     Full name: Mailing List Manager
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\lp (RID: 1014)
|     Full name: lp
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\mail (RID: 1016)
|     Full name: mail
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\man (RID: 1012)
|     Full name: man
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\msfadmin (RID: 3000)
|     Full name: msfadmin,,,
|     Flags: Normal user account
|   METASPLOITABLE\mysql (RID: 1218)
|     Full name: MySQL Server,,,
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\news (RID: 1018)
|     Full name: news
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\nobody (RID: 501)
|     Full name: nobody
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\postfix (RID: 1212)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\postgres (RID: 1216)
|     Full name: PostgreSQL administrator,,,
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\proftpd (RID: 1226)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\proxy (RID: 1026)

```

Findings

The scan revealed:

- Samba server
- Workgroup information
- SMB-related services

Analysis

SMB enumeration can expose usernames, shares, and operating system information that may assist attackers during later attacks.

Challenge 12 — Enum4linux

Objective

To enumerate SMB shares, users, groups, and NetBIOS information.

Command Used

```
enum4linux -a 192.168.56.105
```

Explanation of Command

- enum4linux = Linux enumeration tool for Windows and Samba systems.
- -a = Perform all enumeration techniques.

```
(kali@kali)-[~]
└─$ enum4linux -a 192.168.56.105
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Wed May 6 13:55:22 2026

===== ( Target Information ) =====
Target ..... 192.168.56.105
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.56.105 ) =====
[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.56.105 ) =====
Looking up status of 192.168.56.105
METASPLOITABLE <00> - B <ACTIVE> Workstation Service
METASPLOITABLE <03> - B <ACTIVE> Messenger Service
METASPLOITABLE <20> - B <ACTIVE> File Server Service
.._MSBROWSE_. <01> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1d> - B <ACTIVE> Master Browser
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections

MAC Address = 00-00-00-00-00-00

===== ( Session Check on 192.168.56.105 ) =====
[+] Server 192.168.56.105 allows sessions using username '', password ''

===== ( Getting domain SID for 192.168.56.105 ) =====
Domain Name: WORKGROUP
Domain Sid: (NULL SID)
[+] Can't determine if host is part of domain or part of a workgroup

===== ( OS information on 192.168.56.105 ) =====
```

```
===== ( Users on 192.168.56.105 via RID cycling (RIDS: 500-550,1000-1050) ) =====

[+] Found new SID:
S-1-5-21-1042354039-2475377354-766472396

[+] Enumerating users using SID S-1-5-21-1042354039-2475377354-766472396 and logon username '', password ''

S-1-5-21-1042354039-2475377354-766472396-500 METASPLOITABLE\Administrator (Local User)
S-1-5-21-1042354039-2475377354-766472396-501 METASPLOITABLE\nobody (Local User)
S-1-5-21-1042354039-2475377354-766472396-512 METASPLOITABLE\Domain Admins (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-513 METASPLOITABLE\Domain Users (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-514 METASPLOITABLE\Domain Guests (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1000 METASPLOITABLE\root (Local User)
S-1-5-21-1042354039-2475377354-766472396-1001 METASPLOITABLE\root (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1002 METASPLOITABLE\daemon (Local User)
S-1-5-21-1042354039-2475377354-766472396-1003 METASPLOITABLE\daemon (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1004 METASPLOITABLE\bin (Local User)
S-1-5-21-1042354039-2475377354-766472396-1005 METASPLOITABLE\bin (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1006 METASPLOITABLE\sys (Local User)
S-1-5-21-1042354039-2475377354-766472396-1007 METASPLOITABLE\sys (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1008 METASPLOITABLE\sync (Local User)
S-1-5-21-1042354039-2475377354-766472396-1009 METASPLOITABLE\adm (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1010 METASPLOITABLE\games (Local User)
S-1-5-21-1042354039-2475377354-766472396-1011 METASPLOITABLE\tty (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1012 METASPLOITABLE\man (Local User)
S-1-5-21-1042354039-2475377354-766472396-1013 METASPLOITABLE\disk (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1014 METASPLOITABLE\lp (Local User)
S-1-5-21-1042354039-2475377354-766472396-1015 METASPLOITABLE\lp (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1016 METASPLOITABLE\mail (Local User)
S-1-5-21-1042354039-2475377354-766472396-1017 METASPLOITABLE\mail (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1018 METASPLOITABLE\news (Local User)
S-1-5-21-1042354039-2475377354-766472396-1019 METASPLOITABLE\news (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1020 METASPLOITABLE\uucp (Local User)
S-1-5-21-1042354039-2475377354-766472396-1021 METASPLOITABLE\uucp (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1025 METASPLOITABLE\man (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1026 METASPLOITABLE\proxy (Local User)
S-1-5-21-1042354039-2475377354-766472396-1027 METASPLOITABLE\proxy (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1031 METASPLOITABLE\kmem (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1041 METASPLOITABLE\dialout (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1043 METASPLOITABLE\fax (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1045 METASPLOITABLE\voice (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1049 METASPLOITABLE\cdrom (Domain Group)

===== ( Getting printer info for 192.168.56.105 ) =====

No printers returned.

enum4linux complete on Wed May 6 13:57:00 2026

(kali@kali)-[~]
$
```

Findings

The scan discovered:

- Users
- Groups
- NetBIOS names
- SMB shares
- Workgroup information

Analysis

Enum4linux provides detailed information about SMB configurations and may expose sensitive system information useful for attackers.

Challenge 13 — NFS Exports

Objective

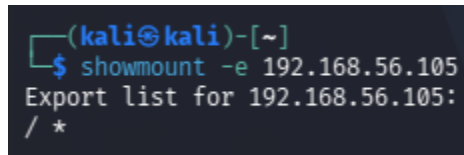
To identify NFS shared directories available on the target machine.

Command Used

`showmount -e 192.168.56.105`

Explanation of Command

- `showmount` = Displays NFS exports.
- `-e` = Show exported directories.



```
(kali㉿kali)-[~]  
$ showmount -e 192.168.56.105  
Export list for 192.168.56.105:  
/ *
```

Findings

The server exported:

`/ *`

Analysis

The target machine exposes NFS shared directories to all hosts. Improperly secured NFS shares may allow unauthorized access to files.

Challenge 16 — Version Detection

Objective

To identify service versions running on the target machine.

Command Used

```
nmap -sV 192.168.56.105
```

Explanation of Command

- -sV = Enable service version detection.

```
(kali@kali)~$ nmap -sV 192.168.56.105
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 13:59 -0400
Nmap scan report for 192.168.56.105
Host is up (0.00067s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:C7:24:F7 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.82 seconds

(kali@kali)~$
```

Findings

Detected services included:

- vsFTPd 2.3.4
- OpenSSH
- Apache HTTP Server
- Samba

Analysis

Version detection helps identify outdated software that may contain vulnerabilities.

Challenge 17 — OS Detection

Objective

To identify the target operating system.

Command Used

```
sudo nmap -O 192.168.56.105
```

Explanation of Command

- -O = Enable operating system detection.

```
(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.56.105
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 14:00 -0400
Nmap scan report for 192.168.56.105
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:C7:24:F7 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds

(kali㉿kali)-[~]
└─$
```

Findings

Nmap identified the target as Linux-based operating system.

Analysis

Operating system detection helps attackers determine suitable exploitation methods.

Challenge 18 — Finger Enumeration

Objective

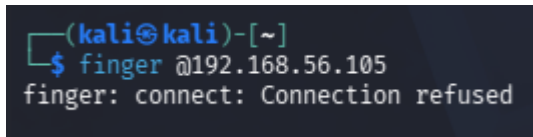
To determine whether the Finger service is enabled on the target machine.

Command Used

`finger @192.168.56.105`

Explanation of Command

- `finger` = Tool used to retrieve user information.
- `@` = Query remote system.



```
(kali㉿kali)-[~]  
$ finger @192.168.56.105  
finger: connect: Connection refused
```

Findings

Output obtained:

`finger: connect: Connection refused`

Analysis

The Finger service is disabled or unavailable on the target machine. This reduces information leakage because attackers cannot enumerate users through the Finger protocol.

Challenge 19 — RPC Information Enumeration

Objective

To enumerate RPC services running on the target machine.

Command Used

```
rpcinfo -p 192.168.56.105
```

Explanation of Command

- `rpcinfo` = Displays RPC service information.
- `-p` = Show port mapping.

```
(kali㉿kali)-[~]  
$ rpcinfo -p 192.168.56.105  
program vers proto  port  service  
100000    2    tcp    111   portmapper  
100000    2    udp    111   portmapper  
100024    1    udp    59620 status  
100024    1    tcp    43074 status  
100003    2    udp    2049  nfs  
100003    3    udp    2049  nfs  
100003    4    udp    2049  nfs  
100021    1    udp    37870 nlockmgr  
100021    3    udp    37870 nlockmgr  
100021    4    udp    37870 nlockmgr  
100003    2    tcp    2049  nfs  
100003    3    tcp    2049  nfs  
100003    4    tcp    2049  nfs  
100021    1    tcp    46561 nlockmgr  
100021    3    tcp    46561 nlockmgr  
100021    4    tcp    46561 nlockmgr  
100005    1    udp    36374 mountd  
100005    1    tcp    34213 mountd  
100005    2    udp    36374 mountd  
100005    2    tcp    34213 mountd  
100005    3    udp    36374 mountd  
100005    3    tcp    34213 mountd  
  
(kali㉿kali)-[~]  
$
```

Findings

The scan identified:

- portmapper
- nfs
- mountd
- status services

Analysis

RPC services provide useful information about NFS and network file-sharing configurations.

Challenge 29 — SMTP Enumeration via Nmap

Objective

To enumerate SMTP users using Nmap scripts.

Command Used

```
nmap -p25 --script=smtp-enum-users 192.168.56.105
```

Explanation of Command

- -p25 = Scan SMTP port.
- --script=smtp-enum-users = Attempt SMTP user enumeration.

```
(kali㉿kali)-[~]  
$ nmap -p25 --script=smtp-enum-users 192.168.56.105  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 14:02 -0400  
Nmap scan report for 192.168.56.105  
Host is up (0.00044s latency).  
  
PORT      STATE SERVICE  
25/tcp    open  smtp  
| smtp-enum-users:  
|_ Method RCPT returned a unhandled status code.  
MAC Address: 08:00:27:C7:24:F7 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds  
  
(kali㉿kali)-[~]  
$
```

Findings

The SMTP service responded successfully and enumeration attempts were performed.

Analysis

SMTP enumeration may reveal valid usernames that attackers can use during brute-force or phishing attacks.

Conclusion

In this lab, multiple enumeration techniques were successfully performed against the target machine. Information gathered included open ports, operating system details, service versions, SMB information, FTP access, NFS exports, and RPC services. Enumeration is an important phase in vulnerability analysis because it helps security professionals identify exposed services and possible security weaknesses.